

Abstract No.

Neural crest-like cells appear in the vascular-microenvironment of glioblastoma at tumor initiation

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The cellular composition and interactions within the glioblastoma (GBM) microenvironment evolve and influence pathology. However, tumor-parenchymal interactions at early stages of tumorigenesis or GBM relapse remain insufficiently characterized. To explore the vascular microenvironment during initiation, compared to tumor-free controls, we developed a methodology in orthotopic GBM mouse models tracing endothelia, mural cells, and neighboring cell types. Lineage analysis in vascular cells was achieved with transgenic strains containing fluorescent reporters. Transfer of fluorescently-labeled, capillary-derived extracellular vesicles (EVs) identified vessel-associated cells in GBM. Dual-fluorescent transgenic GBM models tracking endothelial cells and tumor-associated macrophages displayed EV-transfer over a narrow spatial margin in CNS neoplasms, but not under physiological conditions. Endothelial-derived EV-uptake in other parenchymal populations was observed with immunofluorescence and confocal microscopy. ImageStream-flow-cytometry corroborated this, and traced vascular or EV-labeled vessel-neighboring cells were sorted for single-cell transcriptomics (scRNAseq) at 4 and 14 days. Under physiological conditions, scRNAseq showed that our transgenic lineage-tracing models specifically indicate vascular cells plus smaller macrophage and astrocyte fractions. In GBM tracing models, vessel-associated myeloid cells are increased in the tumor-free hemisphere, with a surge in vessel-associated cells in GBM. Thus, tumor-initiated EV-transfer identified the vascular microenvironment in early and advanced GBM. Strikingly, scRNAseq indicated neural crest-like cells (NCC-like cells), recently discovered in human GBM, within the GBM vascular microenvironment. NCC-like cells are part of the brain injury response to surgical lesions. To determine if NCC-like cells resulted from tumor implantation or initiation, we investigated a transgenic GBM model discriminating these aspects. This model showed that NCC-like cells accumulate in GBM spontaneously. We demonstrated that endothelial-derived vesicle uptake can identify cells associated with the vascular tumor microenvironment even in small tumors. The identification of an early-enriched NCC-like population highlights an underappreciated component of the developing GBM microenvironment and provides a basis to investigate its role in initiation and relapse.



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